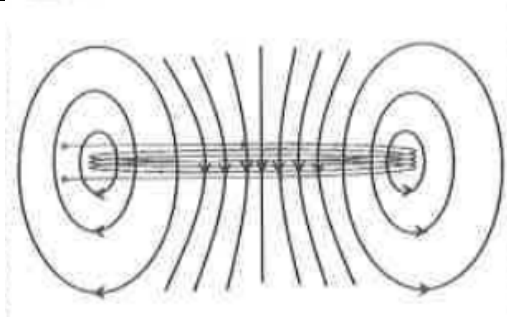


8(a)(i)	Electric field strength at a point is the electric force per unit positive charge placed at that point.
8(a)(ii)	To prevent electrons from colliding with other molecules, as scattered electrons will lose too much kinetic energy, resulting in fewer electrons exiting from the anode with an extremely wide range of kinetic energy.
8(a)(iii)	Gain in Kinetic energy = Loss in electric potential energy $2.48 \times 10^{-16} = e\Delta V$ $\Delta V = \frac{2.48 \times 10^{-16}}{1.60 \times 10^{-19}} = 1550 \text{ V}$
8(a)(iv)	Electrons are emitted from the cathode in different directions. The direction of electric field is horizontal and will only affect the horizontal component of the velocity of the electrons while the vertical component of velocity remained unchanged, hence the resultant velocity will be different.
8(b)(i)	$B = \frac{\mu_o NI}{2r} = \frac{4\pi \times 10^{-7} \times 120 \times 3.5}{2(0.15)} = 0.00176 \text{ T}$
8(b)(ii)	

No.	Solution
8(c)(i)	The velocity of electron is always perpendicular to the magnetic field. By Fleming's left hand rule, a magnetic force always acts perpendicular to both velocity and magnetic field. (i.e. towards a common centre) and the speed of the electron remains unchanged. Hence this magnetic force provides the centripetal force for the electron to move in a circular path.
8(c)(ii)	Magnetic force provide the centripetal force for the electron to move in circular motion. $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ $= \frac{9.11 \times 10^{-31} \times 2.40 \times 10^7}{1.43 \times 0.0017593 \times 1.60 \times 10^{-19}}$ $= 0.054317 = 0.0543 \text{ m}$
8(d)	The velocity of the electron can be resolved into 2 directions: parallel (e.g. $v \cos \theta$) and perpendicular (e.g. $v \sin \theta$) to the magnetic field. By Fleming's left hand rule, a magnetic force always acts perpendicular to both $v \sin \theta$ and magnetic field. This provide centripetal force for the electron to move in a circular path in a plane perpendicular to the magnetic field. As $v \cos \theta$ is parallel to the magnetic field, there will be no magnetic force and the electron will move parallel to the magnetic field. Both motion happened at the same time and hence its moves in a helical path.